

Cyber Security Foundations | CySec

Summary

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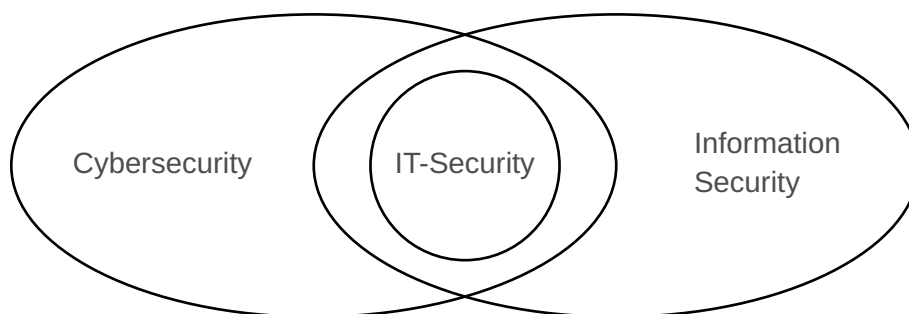
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1. INFORMATION SECURITY

<i>Term</i>	<i>Definition</i>
Information	An organization's data that has been processed, organized, or structured in a way that gives it meaning and value to an organization or individual.
Information security	Protection of the integrity, confidentiality and availability of information data whether in storage, transit or processing.
non-repudiation	prevents parties from denying actions they have performed.
accountability	ability to trace actions and decisions back to a specific person or system.
authentication	verifies the identity of a user or system.
authorization	determines what actions an authenticated entity is allowed to perform.
access control	restricts access to resources based on defined rules.
business continuity	ensures critical operations continue during disruptions.
security policy	a rule or expectation for protecting information.
compliance	adherence to laws, regulations, and (security-)standards.
asset	any item of value belonging to an organization – For example: information, systems, people and processes
attack	an act that intends to damage, steal or degrade an organizations assets
vulnerability	a flaw or weakness in a system that can be abused
exploit	a technique or method used to take advantage of a vulnerability
threat	an event or action with the potential to cause harm by exploiting a vulnerability
risk	the likelihood of a threat exploiting a vulnerability and the potential harm that could cause
control	a measure designed to reduce the potential risk of an attack – Can be achieved through training employees, enforcing policies or implementing technology



1.1. TYPES OF INFORMATION

- Personal information
- Business information
- Financial information
- Intellectual property
 - Copyright
 - Trademarks

- Patents
- Trade secrets
- System information

1.2. HOW CAN INFORMATION BE ATTACKED

- In storage
 - Data that is stored on a server or in a database short-term or long-term.
- In transit
 - Data that is currently being transported from one place to another.
- In use
 - Data that is currently being processed by a service or another entity.

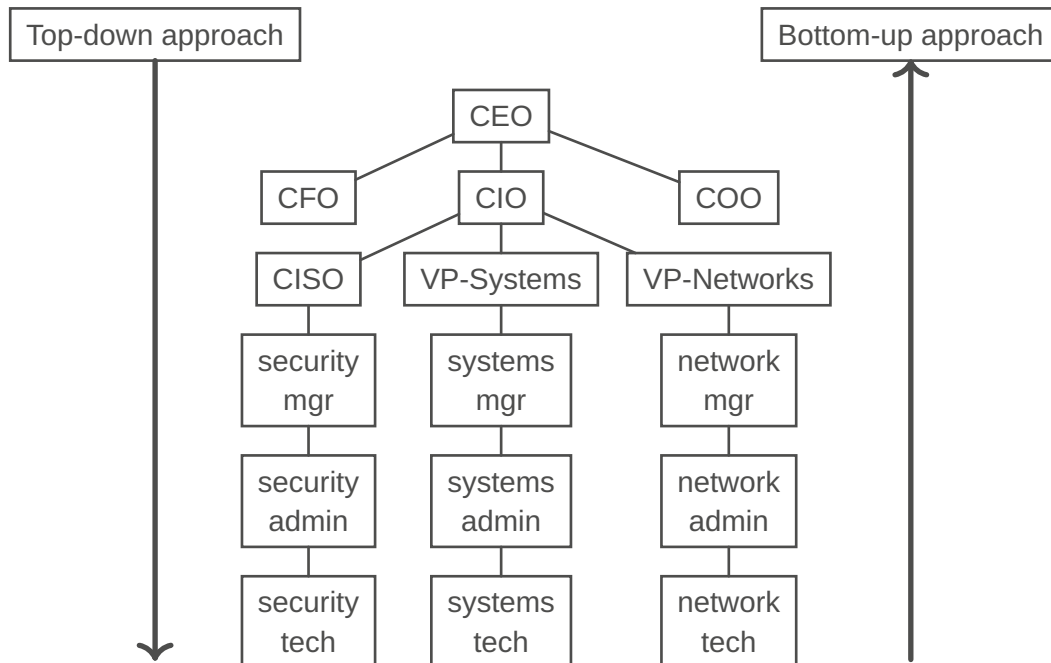
1.3. COMPONENTS OF AN INFORMATION SYSTEM (IS)

- Software
- Hardware
- Data
- People
- Procedures
- Networks

1.4. BALANCING SECURITY AND SYSTEM USABILITY

- Obtaining perfect information security is impossible.
- Security needs to protect the system without slowing people down.
- Too much security can lead to workarounds.
 - Example: If strong passwords are enforced, people might start writing them down on sticky notes.
- Too much convenience exposes the system to unnecessary risks.
- It's all about finding that sweet spot between security and usability.
 - Example Solution: Employees must use multi-factor authentication. This way, they are free to use a less secure password without compromising the overall security.
- An even better, continuously review policies and involve users to find the best solution.

1.5. IMPLEMENTATION OF INFORMATION SECURITY



Bottom-Up

- Initiated by an organization's technical staff (system engineers, admins, etc.).
- Implementations happen before policies are defined.
- Often lacks support from management, budget and consistency.
- Generally less effective and not scalable in large organizations.

Top-Down

- Initiated and supported by an organization's upper management.
- Policies come first and provide guidance for implementations.
- Ensures proper funding, authority and organization-wide enforcement.
- Generally more effective and in-line with the business strategy

1.6. CIA TRIAD

The CIA triad is a foundational information-security model stating that systems should protect:

- **Confidentiality** - Keeping information secret
- **Integrity** - Keeping information correct and unaltered
- **Availability** - Ensuring information and systems remain accessible

	<i>Confidentiality</i>	<i>Integrity</i>	<i>Availability</i>
Goal	Prevent or minimize unauthorized access to information	Protecting the reliability and correctness of information.	Ensuring that subjects have timely and uninterrupted access to information.
Steps to ensure it	– Encryption – Access Control – Allow / enforce advanced authentication mechanisms	– Digital Signatures – Hashing and Checksums – Change Management	– Redundance and Backups – DDoS Protection – Incident Response

1.7. NON-REPUDIATION AND ACCOUNTABILITY

Example of security controls through which non-repudiation can be established: Digital certificates, session identifiers, transaction logs, etc.

1.7.1. Non-Repudiation

- Ensures that the subject of an activity or who caused an event cannot deny having performed an action or cannot deny that the event occurred.
- Non-Repudiation prevents a subject from claiming not to have sent a message, not to have performed an action, or not to have been the cause of an event.

1.7.2. Accountability

- Being responsible or obligated for actions and results.
- Non-Repudiation is an essential part of accountability. A suspect cannot be held accountable if they can repudiate the claim against them.

1.8. STRIDE MODEL

A structured model developed by Microsoft used in cybersecurity to identify and categorize threats to systems by looking at how they can be attacked.

TODO:

Authenticity

Term	Definition
S(poofing)	Pretending to be someone else. (see Authenticity)
T(ampering)	Unauthorized data modification or altering. (see Integrity)
R(epudiation)	Denying actions without proof. (see Non-Repudiation)
I(nformation disclosure)	Exposing sensitive information. (see Confidentiality)
D(enial of service)	Making systems or services unavailable. (see Availability)
E(levation of privilege)	Gaining unauthorized rights or privileges.

1.9. MCCUMBER CUBE

Y-Axis: Security Goals (C.I.A. Triad)

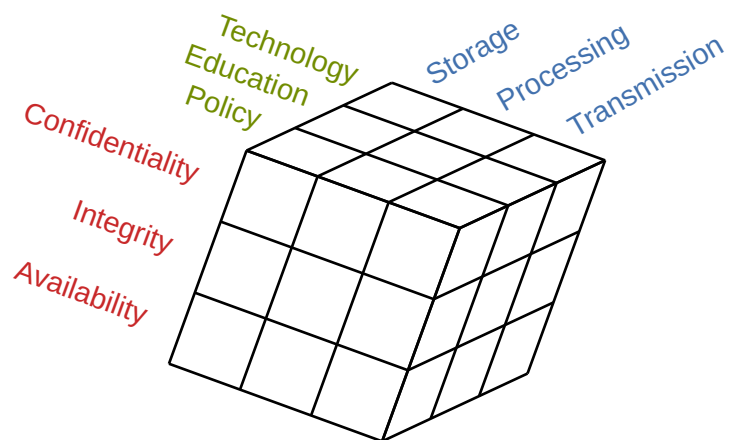
- Defines what needs to be protected.

X-Axis: Information States

- Describes where the information exists.

Z-Axis: Safeguards / Controls

- Defines how protection is implemented.



2. THREAT CATEGORIZATION

<i>Term</i>	<i>Definition</i>
Social Engineering	Manipulating people to reveal confidential information.
Software Attacks	Exploiting vulnerabilities in software to gain access to a system or steal data.
Denial of Service	Overloading one or multiple systems to make it unavailable.
Web Application Attacks	Exploiting vulnerabilities in websites or servers hosting websites.
Password / Authentication Attacks	Attempting to bypass or compromise login systems to gain unauthorized access.
Physical Attacks	Bypassing technical controls by accessing physical infrastructure directly.

2.1. SOCIAL ENGINEERING

The psychological manipulation of individuals to trick them into revealing confidential information or performing actions that can compromise security.

<i>Term</i>	<i>Definition</i>
Phishing	Forged emails impersonating legitimate entities.
Spear Phishing	Targeted phishing against specific individuals.
Vishing	Voice-based phishing over phone or video calls.
Smishing	SMS / Text-based phishing.

2.2. SOFTWARE ATTACKS

Attacks involving malicious code or malware designed to damage systems, steal sensitive data, or gain unauthorized access to systems or services.

<i>Term</i>	<i>Definition</i>
Virus	Malware that attaches to programs and spreads.
Worms	Self-replicating malware that spreads over a network.
Trojan Horse	Malicious software disguised as legitimate applications.
Ransomware	Malware that encrypts victim's data and demands payment to restore access.
Rootkits	Stealthy tools that hide malicious activity and maintain privileged access.

2.3. DENIAL OF SERVICE

Attacks aims at making a system or service unavailable by overwhelming it with excessive traffic or requests.

<i>Term</i>	<i>Definition</i>
DoS	Single source denial of service attacks.
DDoS	Denial-of-service attacks performed by multiple attackers or attacking devices.
Botnet	A network of compromised computers and other devices controlled by an attacker and used to together to flood a target with excessive traffic.

<i>Term</i>	<i>Definition</i>
SYN-Flood Attack	Sending many connection requests without completing them.
Reflection Attack	Attacker sends requests to a service and spoofs the victim's IP making the service send (many) replies to the victim instead of back to the attacker.

2.4. WEB APPLICATION ATTACKS

Exploits vulnerabilities in web applications to steal data, manipulate content, or gain unauthorized access.

<i>Term</i>	<i>Definition</i>
SQL Injection	An attacker inserts malicious SQL commands into an input to manipulate a database and access, modify, or delete data.
Cross-Site Scripting (XSS)	An attacker injects malicious scripts into a website that execute in other users' browsers to steal sensitive data.
Cross-Site Request Forgery (CSRF)	An attacker tricks a logged-in user's browser into sending unauthorized requests to a web application on their behalf.
Broken Authentication	Weak authentication mechanisms allow attackers to compromise passwords, sessions, or identities to gain unauthorized access.

2.5. PASSWORD / AUTHENTICATION ATTACKS

Attacks that attempt to bypass or compromise login systems to gain unauthorized access to a system or service.

<i>Term</i>	<i>Definition</i>
Rainbow Table Attacks	Attackers using precomputed hash lookup tables to reverse weakly hashed passwords back into plaintext.
Password Spraying	Attackers trying a few common password like "password" across many accounts to avoid lockouts or timeouts.
Credential Stuffing	Attackers using leaked usernames and passwords from previous breaches to attempt logins on other services.
Brute Force Attack	Attackers repeatedly try many username and password combinations until they successfully gain access to an account.

2.6. PHYSICAL THREATS

Threats or attacks that affect the physical infrastructure supporting information systems, usually bypassing technical controls overall.

<i>Term</i>	<i>Definition</i>
Theft of devices	Attackers physically steal hardware to gain direct access to stored data, credential, internal systems, or other sensitive data.
Hardware tampering	An attacker modifies or implants malicious components in physical equipment to intercept data, bypass security, or disrupt operations.
Power disruption	Attackers interrupt or manipulate power supply to shut down or destabilize critical systems or services impacting availability and business continuity.

Term	Definition
Environmental damage	Natural or deliberate environmental events that damage infrastructure, causing data loss, downtime, or destruction of critical systems (e.g., earthquake, fire).

3. INFORMATION SECURITY MANAGEMENT

3.1. INFORMATION SECURITY GOVERNANCE

The system by which an organization directs and controls its information security strategy to ensure that it supports business objectives, manages risk appropriately, and complies with legal and other regulatory requirements.

Strategic Direction

- Defining security objectives aligned with business goals.

Leadership and Accountability

- Having clear roles and responsibilities for security decisions.

Risk Management

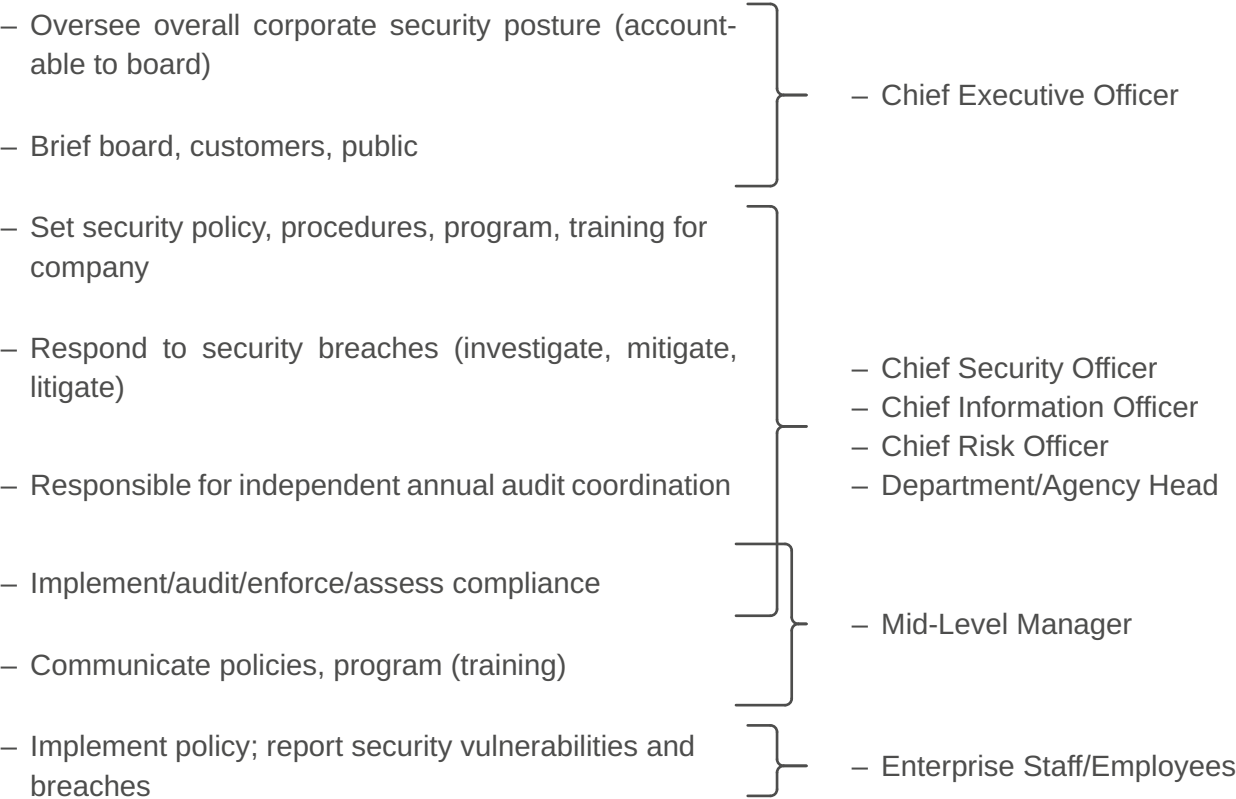
- Defining risks and ensuring they are identified and addressed appropriately.

Regulatory Compliance

- Ensuring adherence to laws and regulations (e.g. NIS2, HIPAA, CRA)

Responsibilities

Functional Role Examples



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3.2. INFORMATION SECURITY MANAGEMENT SYSTEM (ISMS)

A structured framework used to systematically manage and protect an organization's assets through various policies, processes and controls

Security governance defines **what** an organization wants to achieve. An ISMS defines **how** the organization wants to manage it.

Enterprise Information Security Policy (EISP)

- The information security policy that sets the strategic direction and scope for all an organization's security efforts.

Risk Management Process

- Definition of processes to identify assets, analyze threats and evaluate risk.

Security Awareness and Training

- Educational programs to ensure employees understand their security responsibilities.

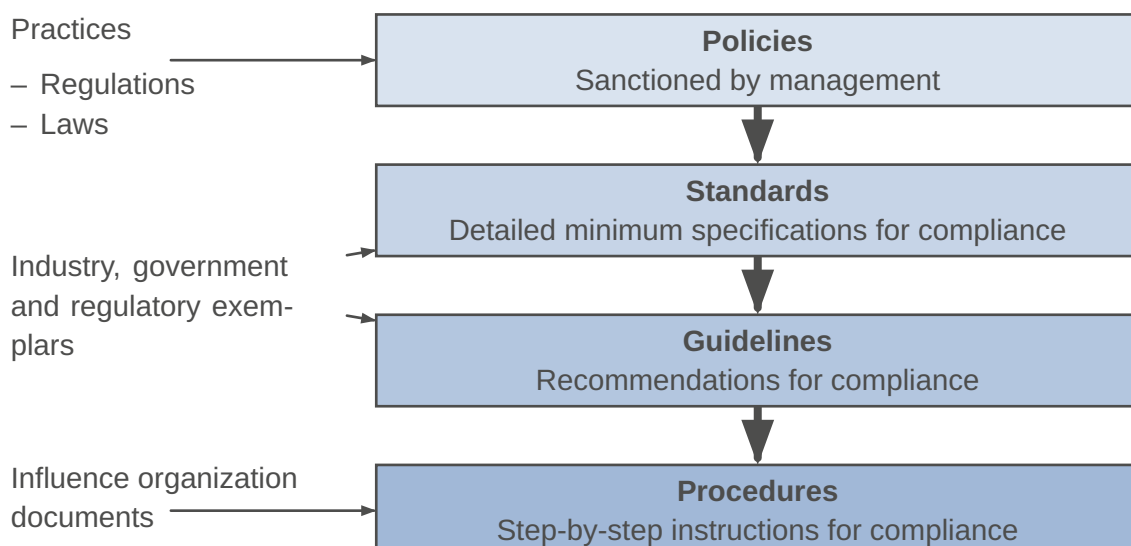
Monitoring, Measurement and Audits

- Ongoing evaluation of control effectiveness and ISMS performance.

4. POLICY

A high-level, management-approved rule that defines mandatory organizational behavior and translates external laws and regulations into enforceable internal requirements.

<i>Term</i>	<i>Definition</i>
policy	instructions that dictate certain behavior within an organization.
guidelines	non-mandatory recommendations employees may use as a reference.
procedures	step-by-step instructions designed to assist employees in following policies.
practices	examples of actions that illustrate compliance with policies.
standard	a detailed statement of what must be done to comply with a policy.
de jure standard	a standard that has been formally evaluated and approved by a formal standards organization
de facto standard	a standard that is widely adopted or accepted by a public group.



What does a policy do? Establishes authority, accountability, and responsibilities for protecting information assets. Provides the foundation for standards, procedures and guidelines.

Who is responsible for policies? Policies are created and approved by senior management, ensuring organizational commitment. Management is responsible for enforcement while employees and users are responsible for compliance.

How is a policy enforced? By clearly communicating it to all relevant parties, integrating it into standards and procedures, monitoring compliance through audits and oversight, and applying defined disciplinary measures when violations occur.

Cyber Resilience Act (EU)

- Requires secure-by-design digital products and vulnerability management (starting December 2027).

Health Insurance Portability and Accountability Act (U.S.)

- Requires administrative, technical, and physical safeguards for protecting patient health data from disclosure.

NIS2 Directive (EU)

- Mandates cybersecurity risk management and incident reporting for critical and important entities.

Local Laws

- Many regions have their own data protection or breach notification laws in addition to national or EU regulations.

4.1. DESIGNING EFFECTIVE POLICIES

1) Development

- Policies must align with organizational goals, business risks and legal requirements.

2) Distribution

- Policies must be distributed to all affected entities in a timely manner.

3) Comprehension

- Policies must be readable for, available to and read by all affected entities.

4) Compliance

- Policies must be formally agreed to by act or affirmation.

5) Enforcement

- Policies must be uniformly applied to all affected entities.

6) Review

- Policies must be reviewed regularly in a changing environment.

4.2. ENTERPRISE INFORMATION SECURITY POLICY (EISP)

The high-level information security policy that sets the strategic direction, scope and tone for all an organization's security efforts and policies.

- Guidance for the development, implementation and management of the security program.
- Sets the requirements that must be met by the information security blueprint.
- Defines the purpose, scope, constraints and applicability of the security program.
- Assigns responsibilities for the various areas of information security.
- Addresses legal compliance.

4.2.1. Elements of an EISP

Although the content of EISP documents varies among organizations, most EISP documents should include the following elements.

- Statement of Purpose
 - Statement of intent that defines the scope, objectives, and purpose of the enterprise information security policy and establishes its role as the foundation for all supporting security documents.
- Information Security Elements
 - Definition of information security that outlines the core principles and concepts, including confidentiality, integrity, and availability, guiding the organization's security efforts.
- Need for Information Security
 - Definition of the importance of information security within an organization and its legal and ethical responsibility to protect information about customers, employees, and markets.
- Information Security Responsibilities and Roles
 - Description of the organizational structure that supports information security, including defined roles and responsibilities for management, employees, and users, as well as responsibility for maintaining the policy itself.

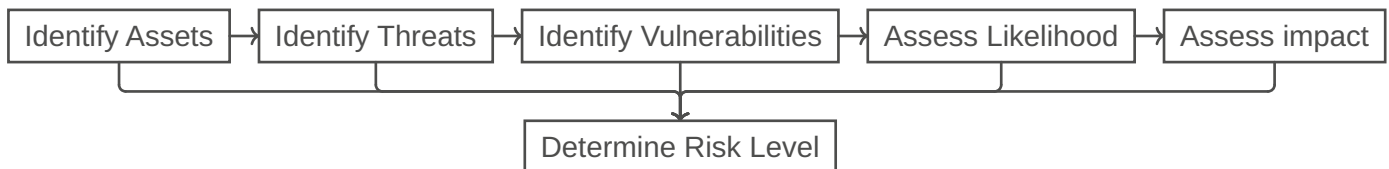
4.3. ISSUE-SPECIFIC SECURITY POLICY

An organizational policy that provides detailed, targeted guidance to instruct members of an organization in the use of a specific resource.

- Supports the EISP by translating it into an issue-specific guidance.
- Establishes rules for access, monitoring, and protection of the resource.
- Defines acceptable and unacceptable use of the specified technology or resource.
- Assigns responsibilities and accountability to users, administrators, and management.

5. RISK ANALYSIS

The process of identifying assets, threats, and vulnerabilities, and evaluating the likelihood and impact of potential adverse events to determine the level of risk.



5.1. IDENTIFYING ASSETS

<i>Term</i>	<i>Definition</i>
Asset	Any resource that has some kind of value to an organization and therefore requires protection.
Information Assets	Customer data, intellectual property, source code, etc.
Technical Assets	Services, applications, databases, networks, etc.
Physical Assets	Servers, devices, facilities, infrastructure, etc.
Human Assets	Employees, administrators, contractors, key personnel, etc.
Business Process Assets	Critical operational workflows

5.2. CLASSIFYING ASSETS

The process of assigning every asset to a class based on their value, sensitivity and impact if compromised

<i>Term</i>	<i>Definition</i>
Public	Information that can be shared without risk
Internal	Information for organization internal use only
Confidential	Sensitive information that could cause harm if disclosed
Restricted	Highly sensitive, strictly limited and strongly protected information

5.3. IDENTIFYING THREATS

A potential event, actor, or action that could exploit a vulnerability and cause harm to an asset.

Examples: Power outage, insider threat, phishing attack

5.4. SECURITY CONTROLS

Measures to reduce risk by detecting, preventing, responding to, or mitigating threats to organizational assets.

TODO:
belongs into information security management

5.4.1. Types

<i>Term</i>	<i>Definition</i>
Administrative / Management Controls	Policies, procedures, security training, security governance, etc.
Technical / Logical Controls	Firewalls, encryption, access control systems, system hardening, etc.
Physical Controls	Physical locks, surveillance cameras, secure access badges, turnstiles, etc.

5.4.2. By Function

<i>Term</i>	<i>Definition</i>
Preventive Controls	Stop incidents before they occur. e.g., Firewalls, access control, encryption, etc.
Detective Controls	Identify incidents when they occur. e.g., Intrusion detection, log monitoring, SIEM, CCTV, etc.
Corrective Controls	Limit damage and restore systems after an incident. e.g., Backups, system restore, incident response, etc.
Deterrent Controls	Discourage malicious behavior. e.g., Warning banners, monitoring notices, disciplinary policies, etc.
Compensating Controls	Reduce risk when a primary control cannot be implemented. e.g., Network isolation, layered security, alternative safeguards, etc.

5.5. BUSINESS CONTINUITY MANAGEMENT

Ensures that critical business functions can continue during and after incidents or disruptions such as cyberattacks, system failures, or physical incidents.

Even with strong security controls in place, incidents can and will still occur at some point. BCM prepares the organization to operate and recover during these times.

5.5.1. Key Objectives

- Maintain critical operations during incidents.
e.g., backups, redundant services, manual processing, etc.
- Minimize downtime and financial impact.
e.g., fast system restore, emergency support contracts, incident response team, etc.
- Protect people, assets and reputation
e.g., evacuation plans, fire suppression systems, customer notification processes, etc.
- Enable fast and structured recovery
e.g., disaster recovery playbooks, tested backup restoration, post-incident review processes, etc.

5.6. SECURITY AND AWARENESS TRAINING

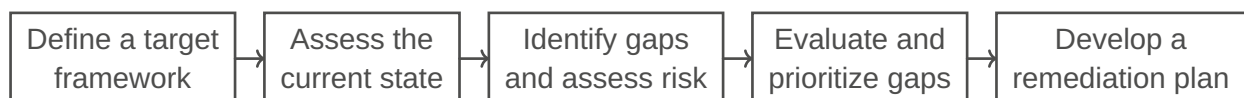
A coordinated program designed to ensure that all members of an organization understand their security responsibilities and have the knowledge and skills to protect information assets.

	<i>Awareness (Level 1)</i>	<i>Training (Level 2)</i>	<i>Education (Level 3)</i>
Objective	Seeks to teach members of an organization what security is and what to do in certain situations	Seeks to train members of an organization how they should react and respond to certain situations	Seeks to educate members of an organization as to why the organization reacts the way it does
Complexity Level	Offers basic information about threats and responses	Offers more detailed knowledge about detecting threats and teaches skills needed for effective reaction	Offers the background and depth of knowledge to gain insight into how processes are developed and enables ongoing improvement
Teaching Method	– Videos – Newsletters – Posters – Informal Training	– Informal Training – Workshops – Hands-on Practice	– Theoretical Instruction – Discussions / Seminars – Background Reading
Impact timeframe	Short-term	Intermediate	Long-term

5.7. GAP ANALYSIS

The process of comparing an organization's current security posture with a required or desired target to identify missing or insufficient controls.

- Risk Analysis: What could go wrong?
- Gap Analysis: Where are we non-compliant or under-protected?



5.8. SECURITY FRAMEWORK

A structured set of principles, processes, and controls that organizations use to manage risks and protect their information systems, assets, and operations.

<i>Framework</i>	<i>Definition</i>
ISO/IEC 27000	Global standard for information security management systems (ISMS).
NIST Cybersecurity Framework	Practical framework for managing cyber risk
CIS Controls	Defines 18 highly practical technical security controls
ISACA CORBIT	An IT governance and risk management framework.

5.9. ISO/IEC 27000

A set of standards for ISMS, helping organizations systematically protect information assets using a risk-based approach.

<i>Standard</i>	<i>Definition</i>
ISO/IEC 27000	Introduction, terminology, and key concepts (e.g., risk, asset, control, etc.)
ISO/IEC 27001	Defines requirements to establish, implement, maintain, and improve an ISMS.
ISO/IEC 27002	Practical guidance for implementing controls.
ISO/IEC 27005	Focuses on risk management methodology.
ISO/IEC 27017	Additional guidance for cloud services.
ISO/IEC 27018	Focuses on privacy and personal data protection in cloud environments.

5.10. NIST CYBERSECURITY FRAMEWORK

A risk-based guideline that helps organizations to structure, manage, and improve their cybersecurity activities across the full lifecycle of prevention, detection, and response.

- It's organized into five core functions; Identify, Protect, Detect, Respond, Recover.
- Provides categories and subcategories of cybersecurity outcomes and controls.
- Includes implementation tiers to assess cybersecurity maturity.
- Is very flexible and adaptable to any organization or business sector.
- Not certifiable, primarily used as guidance and best practice.

5.11. RISK, VULNERABILITIES AND THREATS

5.11.1. Vulnerabilities

A potential weakness in an asset or its defensive control system. Can be **known** or **unknown**.

Examples:

- Software vulnerabilities (Bugs, design flaws, ...).
- Human vulnerabilities (sharing passwords, ...).

5.11.2. Threat

A potential malicious action, or event that aims to damage, or steal unauthorized access to assets. A threat exploits system vulnerabilities.

- Threat: What?
- Threat Actor: Who?
- Threat Vector: How?

5.11.3. Threat Actors

<i>Motivations</i>	<i>Actor Types</i>
– Service disruptions – Data exfiltration – Disinformation – Chaotic / Vandalism – Financial – Blackmailing – Fraud – Political	– Hackers – Unskilled Attackers & Script Kiddies – Hacker Teams & Hacktivists – Nation-State Actors – Organized Crime – Internal Threat Actors

5.11.4. Threat Vector

The path, method, or delivery mechanism that a threat uses to reach an asset and exploit a vulnerability.

Types:

- Software Vectors (Bugs, Virus, ...)
- Network Vectors (Bluetooth, Open Ports, Remote Network, ...)
- Lure-Based Vectors (Drop Attacks with USB Sticks, Trojans, ...)
- Message-Based Vectors (SMS, Email, IM, Web and Social Media, ...)
- Supply Chain Vectors (Updates, Libraries, ...)

5.11.5. Attack surface

The sum of vulnerabilities, pathways, or methods (Threat vectors) that hackers can use to gain unauthorized access to the network or sensitive data, or to carry out a cyberattack.

5.11.6. Risk

The probability of an unwanted occurrence, such as an undesirable event or loss.

- The definition of risk implies threats and vulnerabilities: A risk is only here if we have an existing vulnerability, threat, and threat vector!
- Risk = Vulnerability (Value & Exposure) + Threat (Threat Actor & Threat vector)

At what cost are we willing to accept what risk? The answer to that question gives us risk management.

5.12. RISK MANAGEMENT

The process of identifying, assessing, prioritizing and mitigating threats to an asset from an organisation.

Risk management framework

Structure of the strategic planning and design of the entirety of the risk management efforts (planning).

Risk management process

Implementation, analysis, evaluation of the risk management framework (doing).

TODO:
slides 17

5.12.1. RM Framework

- 1) **Executive Governance & Support:** Support from management and users.
- 2) **Framework Design:** Defining the methods and risk appetite strategy.
- 3) **Framework Implementation:** Rollout of the plan (through → RM process).
- 4) **Monitoring & Review:** How effective is the entire system?

5) **Continuous Improvement**: Continuous adaption to new , or existing threats.

5.12.1.1. Executive Governance and Support

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5.12.1.2. Framework Design

Defining the methods and risk appetite strategy

<i>Term</i>	<i>Definition</i>
Risk appetite (strategic)	The quantity of risk that organizations are willing to accept, to achieve their goals.
Risk tolerance (specific)	The acceptable risk organizations are willing to accept for a specific asset.
Residual risk	The risk that still remains after all controls have been applied.

5.12.1.3. Framework Implementation

Framework Implementation starts after:

- The RM framework and process is finished designing.
- The structure of the RM process & framework is defined.

The methodologies are dependent on the risk appetite:

- Direct rollout
- Pilot-test
- Phased approach

5.12.1.4. Monitoring & Review

- How successful was the framework in the last cycle?
 - Designing
 - Implementing
- What issues require adjustments to the plan?

5.12.2. Risk Management Process

Risk assessment: The identification, analysis, and evaluation of risk as initial parts of risk management.

Risk treatment & Risk Owner: The application of safeguards or controls to reduce the risks to an organization's information assets to an acceptable level.

- 1) **Risk identification**: Where and what is the risk?
- 2) **Risk analysis**: How severe is the current level of risk?
- 3) **Risk evaluation**: Is the current level of risk acceptable?
- 4) **Risk treatment**: What do I need to do to bring the risk to an acceptable level?

5.12.2.1. Risk Identification

The recognition, enumeration, and documentation of risks to an organization's information assets.

Where and what is the risk?

- What are the assets of the organisation? (Internal Asset Register, Weighted Asset Table)
 - Data, Software, Hardware, Networks, Employees, procedures, ...
- What are the threats of the organisation? ([ATT&CK](#) is a globally-accessible knowledge base).
 - Human error, Attacks from hackers, forces of nature, day zero attacks ...
- What are the vulnerabilities? ([CVE](#) and [CVSS](#) helps with that question).

- Lack of training, known bugs in the system, day zero exploit
- Precision is key: If THIS then THAT, because OF ...

5.12.3. CVE & CVSS

- A Common Vulnerabilities and Exposures (CVE) is an industry-wide standard identification number for vulnerabilities.
- The Common Vulnerability Scoring System (CVSS) uses the CIA triad principles within the metrics used to calculate the CVSS base score and assigns severity scores to a vulnerability

5.12.4. Risk Analysis

A determination of the extent to which an organization's information assets are exposed to risk.

Identify the severity of every identified threat and vulnerability.

- What is the probability of an attack?
- What would be the impact of an attack?
 - Quantitative risk analysis assigns real dollar figures to the loss of an asset.
 - Qualitative risk analysis assigns subjective and intangible values to the loss of an asset.
- Existing Security Controls shall be considered

5.12.5. Quantitative Risk Analysis

- 1) Assign Asset Value (AV)
- 2) Calculate Exposure Factor (EF)
- 3) Calculate single loss expectancy (SLE)
- 4) Assess the annualized rate of occurrence (ARO)
- 5) Derive the annualized loss expectancy (ALE)
- 6) Perform cost/benefit analysis of countermeasures

5.12.5.1. AV

- 1) Identify the organization's information assets.
- 2) Classify them.
- 3) Categorize them into useful groups.
- 4) Prioritize them by overall importance.

Example 1 : Weighted Asset Table

		<i>Impact on Revenue</i>	<i>Impact on Profitability</i>	<i>Impact on Reputation</i>		
#		0.3	0.4	0.3	TOTAL (1.0)	Importance
1	Customer order via SSL	5	5	5	5	Critically Important
2	Customer service re- quest via e-mail	3	3	5	3.6	Important
⋮	⋮	⋮	⋮	⋮	⋮	⋮

5.12.5.2. EF

Exposure factor (EF): Represents the percentage of loss that an organization would experience if a specific asset is violated by a realized risk.

- In most cases, a realized risk does not result in the total loss of an asset. The EF simply indicates the expect.

5.12.5.3. SLE

Single loss expectancy (SLE): The cost associated with a single realized risk against a specific asset. It indicates the exact amount of loss an organization would experience if an asset were harmed by a specific threat occurring.

- $SLE = \text{asset value (AV)} \times \text{exposure factor (EF)}$
- Example: if an asset is valued at \$200,000 and it has an EF of 45 % for a specific threat, then the SLE of the threat for that asset is \$90,000.

5.12.5.4. ARO

Annualized rate of occurrence (ARO): The expected frequency with which a specific threat or risk will occur within a single year.

- Example: The ARO of an earthquake in Paris may be .00001, whereas the ARO of an earthquake in San Francisco may be .03 (for a 6.7+ magnitude).

5.12.5.5. ALE

Annualized loss expectancy (ALE): The possible yearly cost of all instances of a specific realized threat against a specific asset.

- $ALE = \text{single loss expectancy (SLE)} \cdot \text{annualized rate of occurrence (ARO)}$
- If the SLE of an asset is \$90,000 and the ARO for a specific threat (such as total power loss) is .5, then the ALE is \$45,000.

On the other hand, if the ARO for a specific threat (such as compromised user account) is 15, then the ALE would be \$1,350,00

5.12.5.6. ALE with Safeguards

- You must calculate the ALE for the asset if the safeguard is implemented.
 - This requires a new EF and ARO specific to the safeguard.
 - The whole point of a safeguard is to reduce the ARO and/or reduce the SLE. The best of all possible safeguards would reduce the ARO to zero.
 - In most cases, the EF to an asset remains the same even with an applied safeguard because if the safeguard fails, the loss on the asset is usually the same as when there is no safeguard.
- Safeguard Costs
 - You must first compile a list of safeguards for each threat. Then you assign each safeguard a deployment value = ACS (Annual cost of the safeguard).

5.12.5.7. Value of a Safeguard

Net Value or Cost/Benefit of a safeguard:

- Negative value: not a responsible choice.
- Positive value: Then the value represents the yearly savings in cost that you CAN have (because the rate of occurrence is just an expected value).

Safety needs to be cost effective. Do not use more resources or money for the protection of an asset as the value of the asset itself!

5.12.6. Risk Evaluation

The process of comparing an information asset's risk rating to the numerical representation of the organization's risk appetite or risk threshold to determine if risk treatment is required.

Risk Evaluation: Compare the risk with the risk appetite of the organization.

- Can the company live with the analysed level of risk (From the CVSS, the quantitative risk analysis, qualitative risk analysis)?
- Levels: Expansionary, Conservative or Neutral

The Risk appetite from the RM Framework must be translated into a value so it can be compared to each analysed risk.

- For the quantitative risk analysis, the risk appetite can be translated into a numerical value!

Goal: The risk must be smaller or equal as the risk appetite.

- Important Indicators for Business Impact:
 - Maximum Tolerable Downtime (MTD)
 - Recovery Point Objective (RPO)
 - Recovery Time Objective (RTO) & Work Recovery Time (WRT)

5.12.7. Risk Treatment

Mitigation risk treatment strategy: The risk treatment strategy that attempts to eliminate or reduce any remaining uncontrolled risk through the application of additional controls and safeguards in an effort to change the likelihood of a successful attack on an information asset; also known as the defense strategy.

The company now has a list of information assets with unacceptable levels of risk.

- The appropriate strategy must be selected and applied.

Four basic strategies to treat risk:

- 1) Mitigation: Apply safeguards that eliminate or reduce the remaining uncontrolled risk.
 - Example: Firewall, Training, ...
- 2) Transfer: Shift risks to other areas or outside entities.
 - Example: Outsourcing
- 3) Acceptance: Understand the consequences of choosing to leave an information assets vulnerability facing the current risk level (after formal evaluation).
- 4) Termination: Remove or discontinue the asset from the organization's operating environment.

5.12.7.1. Mitigation

- Fix vulnerabilities
- Applying controls (tools, processes, rules to mitigate risk)
 - Endpoint Hardening (preventive Control): Secure a “endpoint” (device: laptop, server, ...) by reducing its vulnerabilities and shut down potential threat vectors!
- Reduce final impact (If zero-day attacks, unknown vulnerabilities, or a taken risk happen)
 - EDR (Endpoint Detection and Response): Software that watches for suspicious behaviour and responds with certain measures.
 - XDR (Extended Detection and Response): Watching everywhere (not just on endpoints) and respond with certain measures (shut down infected laptop, ...)

5.12.8. Other RM Frameworks

- OCTAVE (Operationally Critical Threat, Asset and Vulnerability Evaluation) by the Carnegie Mellon University.
- FAIR (Factor Analysis of Information Risk) by Jack A. Jones.
- ISO Standards: ISO 27005 and ISO 31000: (explanation: https://en.wikipedia.org/wiki/ISO/IEC_27005).
- NIST Risk Management Framework (RMF): <https://csrc.nist.gov/publications/sp>

6. IDENTITY & ACCESS MANAGEMENT (IAM)

IAM deals with provisioning and protecting digital identities and user access permissions. Or in other words: The right people can access the right resources for the right reasons at the right time. To ensure this we need Access Controls.

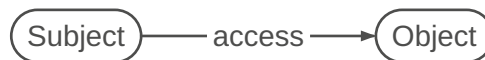
6.1. ACCESS CONTROL

Any hardware, software, or administrative policy or procedure that controls access to resources. The selective method by which systems specify who may use a particular resource and how they may use it.

The goal is to:

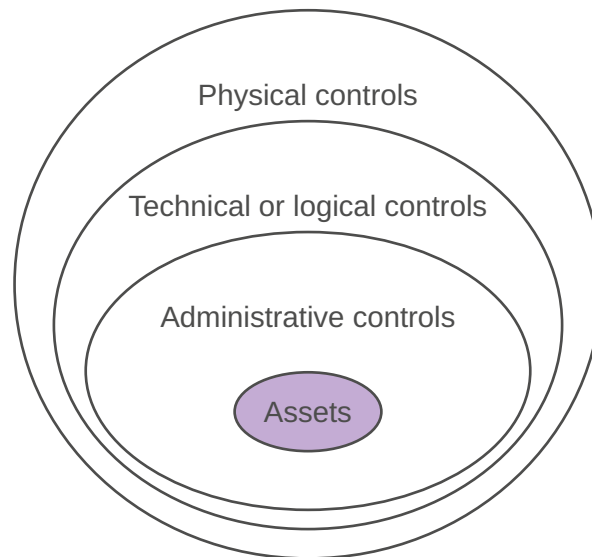
- PROVIDE access to authorized subjects
- PREVENT access to unauthorized access attempts and unauthorized subjects

Term	Definition
Subject	Active entity that accesses a passive object. – Anything that can access a resource can be a subject: users, programs, processes, services, computers,...
Object	Passive entity that provides information to subjects – Anything that can provide resources: files, databases, computers, programs, processes, services, printers, ...



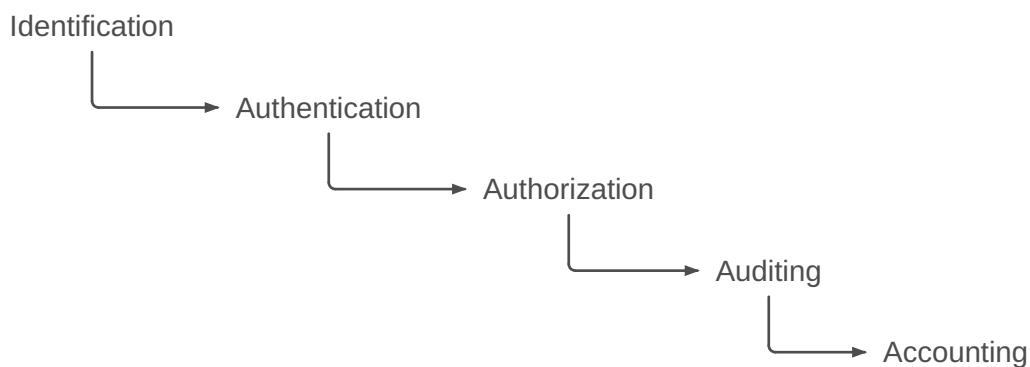
6.1.1. Control methods

Term	Definition
Physical controls	Items that you can physically touch. Included are physical mechanisms deployed to prevent, monitor, or detect direct contact with systems or areas within a facility – Examples: guards, fences, motion detectors, locked doors, sealed windows, lights, cable protection, laptop locks, badges, swipe cards, guard dogs, video cameras, mantraps, and alarms
Technical or logical controls	Hardware or software mechanisms used to manage access and to provide protection for resources and systems – Examples: authentication methods (username, passwords, biometrics,...), encryption, access control lists, protocols,...
Administrative controls	Policies and procedures defined by an organization's security policy or other regulations or requirements – Examples: policies, procedures, hiring practices, background checks, data classifications and labeling, security training,...



TODO:
slides 9-10

6.2. MECHANISMS



6.2.1. Identification

The subject is claiming an identity.

- Example: Typing a username, swiping a smartcard, waving a token device, speaking a phrase, or positioning your face, hand, or finger in front of a camera or in proximity to a scanning device

Important: All subjects must have unique identities

- IT systems track activity by identities, not by the subjects themselves
- A subject's identity is typically labeled as, or considered to be, public information

A subject must provide an identity to a system to start the other processes (authentication, authorization, and accountability)

6.2.2. Authentication

The process of verifying that the claimed identity (from identification) is valid

- Example: password
- Identification and authentication are often used together as a single two-step process

Authentication information used to verify identity is private information and needs to be protected

To authenticate the claimed identity it is common to use multiple factors. These factors are often categorized in three different categories:

- 1) Something you know
 - A Type 1 authentication factor is something you know. Passwords, PINs, ...
- 2) Something you have
 - A Type 2 authentication factor is something you have. Physical devices that a user possesses can help them provide authentication
- 3) Something you are / you do
 - A Type 3 authentication factor is something you are or something you do. It is a physical characteristic of a person identified with different types of biometrics

6.2.2.1. Authentication Schemes

- 1) Basic Authentication: Classical username / password pair transmitted in the clear
- 2) One Time Passwords: Transmitted in the clear but used only once
- 3) Challenge / Response: Response is a function of password and one-time challenge
- 4) Anonymous Key Exchange: Exchange credentials over unauthenticated secure channel
- 5) Zero-Knowledge Password Proofs: Does not permit offline-based password attacks
- 6) Server Certificates plus User Authentication: Transmit user password over unilaterally authenticated secure channel
- 7) Mutual Public Key Authentication: Bilateral use of public key signatures

Attack vulnerability Matrix

Attack	1	2	3	4	5	6	7
Passive Password Sniffing	x						
Offline Brute Force Password Attack	x		x	x			
Active Man-in-the-Middle Attack	x	x	x	x			
Identity Theft on Server	x	x	x	x	x	x	
CA Compromise						x	x

6.2.2.2. Type 1 Factor: Passwords

Passwords are typically static. They are the weakest form of authentication

- Users often choose passwords that are easy to remember and therefore easy to guess or crack
- Randomly generated passwords are hard to remember, and many users write them down
- Users often share their passwords, or forget them
- Passwords are rarely stored in plaintext.
 - A system will create a hash of a password using a hashing algorithm
- Best practices and policies
 - Enforce a minimum length
 - Complexity rules (uppercase/lowercase, non-alphanumeric, etc...)
 - Ageing and expiration
 - Reuse and history
- Password managers mitigate the risk of poor credential management

6.2.2.3. Type 2 Factor: Tokens

A token device, or hardware token, is a device that users can carry with them

- An authentication server stores the details of the token, so at any moment, the server knows what number is displayed on the user's token

Hard Authentication Tokens

- No transmission of the token itself e.g. Smartcards, Hardware OTP Token

Soft Authentication Tokens

- Software token transmitted to the user e.g. via Authenticator App, SMS, Email or phone

Dynamic Password Tokens

- Synchronous create synchronous dynamic passwords are synchronized with an authentication server
- Asynchronous asynchronous dynamic password is based on a Challenge-Response principle

6.2.2.4. Type 2 Factor: Smartcard

A smartcard is a credit card-sized ID or badge and has an integrated circuit chip embedded in it

- Smartcards store information about the authorized user that is used for identification and/or authentication purposes
- Implements certificate-based authentication (private key and sometimes a PIN to activate the card)
- Most current smartcards include a microprocessor and one or more certificates. The certificates are used for asymmetric cryptography such as encrypting data or digitally signing email
- Smartcards are tamper-resistant and provide users with an easy way to carry and use complex encryption keys

6.2.2.5. Type 2 Factor: One-Time Passwords

Onetime passwords are dynamic passwords that change every time they are used

- Onetime password generators are token devices that create passwords
- The PIN can be provided via a software application running on the user's device (e.g., smartphone)

TOTP (Time-based One-Time Password)

- Uses a timestamp and remains valid for a certain timeframe, such as 30 seconds
- This is similar to the synchronous dynamic passwords used by tokens

HOTP (HMAC-based One-Time Password)

- Includes a hash function to create onetime passwords. It creates HOTP values of six to eight numbers
- This is similar to the asynchronous dynamic passwords created by tokens. The HOTP value remains valid until used

6.2.2.6. Type 3 Factor: Biometrics

Biometric authentication uses physiological characteristics to provide authentication for a provided identification.

Errors: Biometrics make measurements and compare them with unique points of reference. This leads to errors (measurements always have errors):

- False reject rate (Type 1 Error): percentage of authorized users who are denied access
- False accept rate (Type 2 Error): percentage of unauthorized users who are granted access
- Crossover error rate (CER): The point at which the rate of false rejections equals the rate of false acceptances

6.2.2.7. Multifactor Authentication

Multifactor authentication is any authentication using two or more factors

- For a positive authentication, elements from at least two, and preferably three factors should be verified
 - When two authentication methods of the same factor are used together, the strength of the authentication is no greater than it would be if just one method were used
 - Using more types or factors results in more secure authentication

TODO:

comparison (slides 24)

6.2.2.8. Secondary Authentication Factors

In addition to the three primary authentication factors, there are some others

- Somewhere You Are
 - The somewhere-you-are factor identifies a subject's location based on a specific computer, a geographic location identified by an Internet Protocol (IP) address, or a phone number identified by caller ID
- Somewhere You Aren't
 - Many IAM systems use geolocation technologies to identify suspicious activity
 - For example, imagine that a user typically logs on with an IP address in Switzerland. If a user is trying to log on from a location in India, it can block the access even if the user has the correct username and password

6.2.2.9. Authentication Frameworks

- Kerberos: Create Authentication through a trusted third party.
- RADIUS: Provide centralized authentication, authorization, and accounting (AAA) for network access.

6.2.2.9.1. Kerberos

An authentication system that uses symmetric key encryption to validate an individual user's access to various network resources by keeping a database containing the private keys of clients and servers that are in the authentication domain it supervises.

- Authentication in UNIX-based TCP/IP networks
- Use of symmetrical cryptography (DES)
- Relies on the mediation services of a trusted referee or notary
- Based on the work by Needham and Schroeder on trusted third-party protocols as well as Denning and Sacco's modifications of these
- Current release is Kerberos v5 ([RFC 1510](#) , September 1993)
- V5 supports additional encryption ciphers besides DES

<i>Term</i>	<i>Definition</i>
Principal	A Kerberos participant
Principal's Master Key (MKey _p)	TODO:
Kerberos Ticket	TODO:
Authentication Server (AS)	Verifies who the client is, gives TGT
Ticket Granting Server (TGS)	Grants access to specific services, gives ST
Ticket Granting Ticket (TGT)	Given by the AS
Service Ticket (ST)	Given by the TGS
Key Distribution Center (KDC)	TODO:

Kerberos Step-By-Step

- 1) The user wants to get authenticated at a Service.
- 2) The user sends a request to the Authentication Server (KDC) asking for a Ticket Granting Ticket (TGT). This request is encrypted with the hash of the user's password.

- 3) The Authentication Server looks up the user, authenticates him using the hashed password and sends back the TGT. (notice, the password itself never travels across the network)
- 4) The user wants to access a specific service. He sends the TGT to the Ticket Granting Server (TGS)
- 5) The TGS verifies the TGT and issues a Service Ticket to the client
- 6) The client presents this Service Ticket directly to the Service he wants to use.
- 7) The Service decrypts the ticket, verifies the client and grants access. He can also send a message back to the client to prove its own identity.

TODO:

diagrams (slides 20,28)

6.2.2.9.2. Remote Authentication Dial-In User Service (RADIUS)

A networking protocol that provides centralized Authentication, Authorization and Accounting (AAA) management for users who use a network service.

Used to secure network nodes: Enterprise Wi-Fi (802.1x), VPNs, Switches

AAA

- Authentication: Verifying the user's identity
- Authorization: Granting specific network privileges (assigning specific IP, ...)
- Accounting: Tracking network resource for auditing, billing, ...

RADIUS Architecture

- 1) User requests network access from the NAS
- 2) NAS prompts the RADIUS server for credentials (username / password, or certificate)
- 3) RADIUS server evaluates the request and returns one of three responses:
 - Access-Accept: User is authenticated, NAS grants network access
 - Access-Reject: Invalid credentials, NAS denies access
 - Access-Challenge: Server requires more information (MFA, or Token)
- 4) When connected, NAS sends Accounting-Request to log the session.

6.2.2.9.3. RADIUS vs Kerberos Vulnerability Matrix

<i>Attack</i>	<i>Kerberos</i>	<i>RADIUS</i>
Passive Password Sniffing		
Offline Brute Force Password Attack	×	×
Active Man-in-the-Middle Attack		×
Identity Theft on Server		
CA Compromise		×

6.2.3. Authorization

The process of authorization ensures that the requested activity or access to an object is possible given the rights and privileges assigned to the authenticated identity

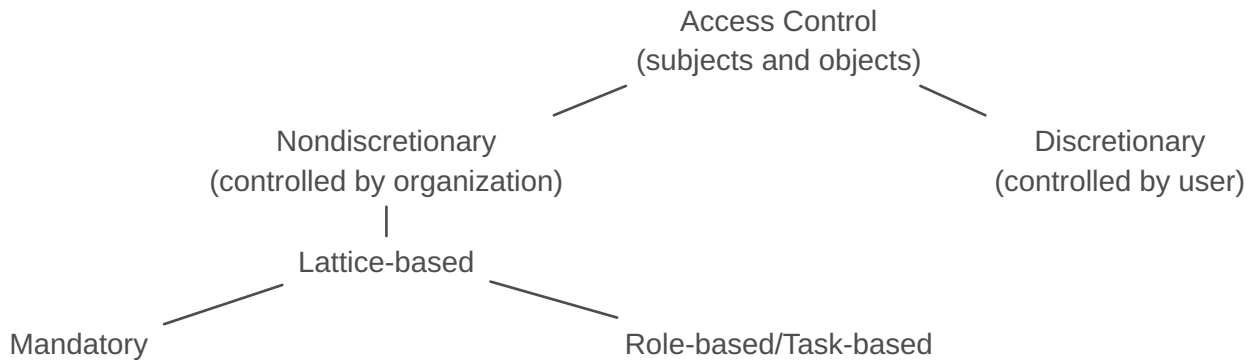
Or in other words: Once a subject is authenticated, access must be authorized

- Just because a subject has been identified and authenticated does not mean they have been authorized to perform any function or access all resources within the controlled environment

Identification and authentication are all-or-nothing aspects of access control. This is NOT the case with authorization:

- Authorization has a wide range of variations between all or nothing for each object within the environment

6.2.3.1. DAC and NDAC



6.2.3.1.1. Discretionary access control (DAC)

Access controls that are implemented at the judgment or option of the data owner. Every object has an owner, and the owner can grant or deny access to any other subjects → The owner (or user) chooses who has access!

- Most flexible and widely used e.g. file system security
- Data owner can modify access control list (ACL)
- Example: User has a hard drive and wants to share it with coworkers. He decides who he shares it with.

6.2.3.1.2. Nondiscretionary access control (NDAC)

Access controls that are implemented by a central authority.

- Example: US-Hospital where access is based on rules and regulations like HIPAA (DSG covers that in Switzerland)

6.2.3.1.2.1. Lattice-based access control (LBAC)

A variation on mandatory access controls that assigns users a matrix of authorizations for particular areas of access, incorporating the information assets of subjects such as users and objects.

- Mandatory access control (MAC): Use of labels applied to both subjects and objects. This means each collection of information is rated, and all users are rated to specify the level of access.
 - Example: Information are labelled as top secret → only users that are labelled top secret are granted access to this information!
- Role-based (RBAC) / Task-based (TBAC) access control: privileges are tied to a role or a job (role-based) or to a task or assignment (task-based).
 - Example: Project manager has access to corresponding information about his project. (role-based)
 - Example: A technician is only allowed into a server room in his planned maintenance timeslot (task-based)

6.2.3.2. Least privilege design principle

Access rights should be limited in scope, time, and function

- “Just enough access” is usually better than broad permanent access

Users and systems should only get the permissions they actually need

- Reduces attack surface and limits damage after account compromise
- Helps prevent misuse of admin accounts and service accounts
- Supports separation of duties and stronger compliance

Requires regular access reviews and removal of unused permissions

6.2.4. Auditing

A subject's actions are tracked and recorded

Purpose: Hold the subjects accountable for their actions while authenticated on a system

6.2.5. Accounting

The consumption of resources by a subject is measured, metered, and collected.

Purpose: Provide a record of resource usage for billing, capacity planning, and trend analysis.

6.3. ESTABLISHING ACCOUNTABILITY AND NON-REPUDIATION

Accountability means actions can be traced to a specific identity

- Proving to regulators that your data is secure
- Link a human to the activities of an identity
- Requires unique user identities, no shared accounts and strong authentication
- Support your security decisions and their implementation
- Supports incident investigation, compliance, and trust in transactions

Non-repudiation means a user cannot credibly deny a performed action

- Logging and audit trails must be complete, accurate, and protected
- Digital signatures are a key mechanism for non-repudiation

6.4. COMMON ACCESS CONTROL ATTACKS

6.4.1. Access Aggregation Attacks (passive attack)

- Access aggregation refers to collecting multiple pieces of nonsensitive information and aggregating them to learn sensitive information.
- Reconnaissance attacks are access aggregation attacks that combine multiple tools to identify multiple elements of a system, such as Internet Protocol (IP) addresses, open ports, running services, operating systems.

6.4.2. Password Attacks (brute-force attack)

- Online: Attacks against online accounts
- Offline: to steal an account database and then crack the passwords.

6.4.3. Dictionary Attack (brute-force attack)

An attempt to discover passwords by using every possible password in a predefined database or list of common or expected passwords also called a password-cracking dictionaries

- Dictionary attack databases also include character combinations commonly used as passwords, but not found in dictionaries
- Dictionary attacks often scan for one-upped-constructed passwords. A one-upped-constructed password is a previously used password, but with one character different.
- For example, password1 is one-upped from password, as are Password, 1password, and passXword

6.4.4. Birthday Attack (brute-force attack)

A birthday attack focuses on finding collisions. Its name comes from a statistical phenomenon known as the birthday paradox

- The birthday paradox states that if there are 23 people in a room, there is a 50 percent chance that any two of them will have the same birthday. (This is not the same year, but instead the same month and day, such as March 30)

- With February 29 in a leap year, there are only 366 possible days in a year. With 367 people in a room, you have a 99.99 percent chance of getting at least two people with the same birthdays. Reduce this to only 23 people in the room, and you still have a 50 percent chance that any two have the same birthday

You can reduce the success of birthday attacks by using hashing algorithms with enough bits to make collisions computationally infeasible, and by using salts.

- MD5 is not collision free
- SHA-3 (short for Secure Hash Algorithm version 3) can use as many as 512 bits and is considered safe against birthday attacks and collisions – at least for now

6.4.5. Rainbow Table Attacks

A rainbow table reduces the time by using large databases of precomputed hashes

- It takes a long time to find a password by guessing it, hashing it, and then comparing it with a valid password hash

A password cracker can then compare every hash in the rainbow table against the hash in a stolen password database file

- When using the rainbow table, the password cracker doesn't spend any time guessing and calculating hashes. It simply compares the hashes until it finds a match
- This can significantly reduce the time it takes to crack a password

Salting

- adds a unique random value to each password before hashing
- prevents identical passwords from producing identical hash values

6.4.6. Sniffer Attacks

A sniffer (also called a packet analyzer or protocol analyzer) is a software application that captures traffic traveling over the network

- A sniffer attack (also called eavesdropping attack) occurs when an attacker uses a sniffer to capture information transmitted over a network

The following techniques can prevent successful sniffing attacks:

- Encrypt all sensitive data (including passwords) sent over a network. Attackers cannot easily read encrypted data with a sniffer
- Use onetime passwords (OTP) when encryption is not possible or feasible. OTPs prevent the success of sniffing attacks, because they are used only once, also see next chapter Kerberos
- Protect network devices with physical security. Controlling physical access to routers and switches prevents attackers from installing sniffers on these devices

7. CRYPTOGRAPHY

<i>Term</i>	<i>Definition</i>
Plaintext/Cleartext	Before a message is put into a coded form, it is known as a plaintext or cleartext
Ciphertext/Cryptogram	The sender of a message uses a cryptographic algorithm to encrypt the plaintext and produce a ciphertext or cryptogram
Cipher	Encryption algorithm An algorithm is a set of rules, usually mathematical, that dictates how enciphering and deciphering processes are to take place
Key/Cryptovvariable	A key is nothing more than a number (usually a very large binary number)

Term	Definition
Key space	– Every algorithm has a specific key space. A key space is defined by its bit size – The key space is the range of numbers from 0 to 2^n, where n is the bit size of the key – A 128-bit key can have a value from 0 to 2^{128}
One-Way Functions	A one-way function is a mathematical operation that easily produces output values for each possible combination of inputs but makes it impossible to retrieve the input values – In practice, it's never been proven that any specific known function is truly one way – Cryptographers rely on functions that they believe are one way – It's always possible that they might be broken by future cryptanalysts
Reversability	Being able to undo the operation of encryption
Nonce	The nonce must be a unique number each time it is used – Used to make sure that a key is not re-used twice – The main feature of a nonce is that it is a number that is only used once (nonce) – It can be a counter (for example) – The nonce is public, whereas the (shared) key is private
Initialization vector (IV)	An IV is a random bit string – It is the same length as the block size and is XORed ($\oplus$) with the message – IVs are used to create unique ciphertext every time the same message is encrypted using the same key

7.1. OBJECTIVES

- Confidentiality (Privacy)
 - Only authorized persons should read a message, get to know sender/receiver, know about the existence of a message
- Integrity
 - Data should be demonstrably unaltered from sender to recipient
- Authentication
 - The identity of an author of a message should be clearly verifiable
- Non-repudiation
 - The sender of a message should not be able to deny authorship or having performed an action.
- Procedures do not necessarily have to fulfill every objective

7.2. KERCKHOFF'S PRINCIPLES

Term	Definition
Security through obscurity	The security of a system or process depends on the confidentiality of its secrecy of its functioning
Kerckhoff's Principle (Auguste Kerckhoffs, 1883)	The security of an encryption method is based on the secrecy of the key and not on the secrecy of the encryption algorithm

A cryptographic system should be secure even if everything about the system, except the key, is public knowledge

- This principle makes algorithms known and public, allowing anyone to examine and test them
- The principle can be summed up as “The enemy knows the system”
- Public exposure may expose weaknesses more quickly, leading to the abandonment of insufficiently strong algorithms and quicker adoption of suitable ones
- A large number of cryptographers adhere to this principle, but not all agree
- Some believe that better overall security can be maintained by keeping both the algorithm and the key private

7.3. SHANNON’S PRINCIPLES

Term	Definition
Confusion	Confusion occurs when the relationship between the plaintext and the key is so complicated that an attacker can’t merely continue altering the plaintext and analyzing the resulting ciphertext to determine the key – The mapping between input and output is very confusing – Substitution of bytes adds confusion – Example: S-Box
Diffusion	Diffusion occurs when a change in the plaintext results in multiple changes spread throughout the ciphertext – A small change in the input leads to a big change on the output – Permutation of bytes adds diffusion; also known as a transposition – Example: P-Box

7.4. CAESAR CIPHER

One of the earliest known cipher systems was used by Julius Caesar to communicate with Cicero in Rome while he was conquering Europe

- To encrypt a message, you simply shift each letter of the alphabet three places to the right
- The Caesar cipher became known as the ROT3 (or Rotate 3) cipher
- The Caesar cipher is a substitution cipher that is mono-alphabetic

7.5. SP-NETWORK

An SP-Network (Substitution-Permutation Network) is an algorithm that uses repeated substitution and permutation operations

- Substitution: Replacing bytes with others
- Permutation: Swapping bytes around
- The substitutions and permutations are combined into a round.
- Rounds are then repeated many times

7.6. XOR

XOR is a binary operator between two values that returns true if either input or the other is true but not both

- Extremely useful in cryptography
- x is deciding whether y will change
- Applying XOR twice reverses its effect
 - Think of x as encrypting y , and then decrypting it again

x	y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0

7.7. ONE-TIME PAD (OTP)

We can design a cipher that uses XOR to encrypt and decrypt a message

- Use a key that's the same length as the message
- XOR each message bit with each key bit

If you take away the key, there is no way to find the message because there is no statical mapping between the input and the output

```

M 01011010 00110101
      ^
K 01001011 10111001
      =
C 00010001 10001100

```

But

- The OTP is not practical
- A 1 GB file would need a 1 GB key!
- If you ever reuse a key, the entire cipher is broken

7.8. HASHING

- A hash function maps data of any size to a fixed-size output in a deterministic and hard-to-reverse way
- A 128 bit hash function will return a 128 bit string, regardless of how much bits have been hashed in
- Hash functions are used everywhere. Message authentication, integrity, passwords etc.
 - For example, SHA-256 can be used to verify data integrity
- A good hash algorithm should perform quickly but it shouldn't be too quick because if it's too quick, it is easy to break

7.8.1. How it works

- Usually hash functions iteratively jumble blocks of a message after another
 - This is a one-way function
 - There is no way to revert back and restore the initial message
- A hash function is kind of like a washing machine for bits
 - The initial hash is usually defined in the spec
 - A new current hash is created every round
 - We loop for every block of the message
- When we run out of message, we use the current hash as the final hash

7.8.2. Strong hash functions

The output must be indistinguishable from random noise

- It should look like you have just generated random numbers
- It shouldn't look like the output is based on the input
- With SHA-256, the output is pretty much undistinguishable from random noise

Bit changes must diffuse through the entire output

- This is called the avalanche effect
- With SHA-256, a small change in the message makes a big change in the hash

7.8.3. Important properties

It shall to be quick but not too quick

It shall introduce diffusion

- Most hash functions, even MD5, adhere to that

Given a hash, we can't reverse it

- Most hash functions, even MD5, adhere to that

Given a message and its hash, we can't find another message that hashes to the same thing

- That's a collision
- That is problematic because we use hashes to verify that things hasn't been changed. So finding collision undermines the whole idea
- MD5 is broken in that sense
- shattered.io: two different PDFs that have the exact same SHA-1 hash

7.8.4. Current standards in cryptographic hash functions

The current standard is the SHA-2 family with 256-bit and 512-bit variants

SHA-3 is not better or worse than SHA-2

- SHA-3 is a completely different function (Keccak algorithm)
- SHA-3 was designed in case something happens to SHA-2

7.8.4.1. KMAC 128/256

KMAC 128/256 is a new SHA-3 based KECCAK MAC

- standardized in Dec 2016, NIST SP 800-185
- alternative method to SHA-2
- permutation-based hash algorithm (sponge construction)
- very strong resistance to the pre-image

Optimized for parallel processing; efficient on various platforms

7.8.5. Hash functions for Password Storage

Hash functions such as SHA-256 are not good to store passwords because there are too fast

- They are designed to be quick to summarize data
- Vulnerable to brute-force attacks. The attackers hash the passwords and compare with the hashes to see if there are any matches

The hashes are iterated to slow them down on purpose

- Argon2 is memory-hard and designed to resist brute-force attacks
- Configurable parameters for memory usage, iterations, and parallelism.
- Slows down attackers – makes massive guessing attempts significantly harder.
- This is exclusively good for login and passwords and not suitable for general-purpose hashing
- It is totally useless for any kind of other hash function usage

7.8.6. Comparison of hash families

Type/ Family	Output Length	Rounds	Security	Use / Application	Examples / Libraries
MD5	128-bit	4	128-bit, fast, insecure	Legacy systems, checksums	OpenSSL, hashlib (Python)
SHA-1	160	80	160-bit, insecure	Legacy signatures, integrity checks	OpenSSL, hashlib (Python)
SHA-2	224, 256, 384, 512	64, 80	Secure, widely used	Digital signatures, certificates	OpenSSL, hashlib (Python)
SHA-3 (Keccak)	224, 256, 384, 512	24	Resistant to certain attacks, flexible	Modern crypto applications	hashlib (Python ≥3.6)
bcrypt	184-bit		Adaptive, salted, GPU-resistant	Password storage	bcrypt (Python)

<i>Type/ Family</i>	<i>Output Length</i>	<i>Rounds</i>	<i>Security</i>	<i>Use / Application</i>	<i>Examples / Libraries</i>
Argon2	256		Winner of Password Hashing Competition, highly secure, configurable	Password storage, key derivation function	argon2-cffi (Python)
PBKDF2	Varies		Iterative, widely supported, configurable iterations	Password storage	hashlib.pbkdf2_hmac (Python)

7.8.7. Where are hashes used?

Hashing lets us ensure that a message hasn't been altered

- 1) Digital signatures
- 2) Message Authentication Codes (MAC)

7.8.8. HMAC

- MAC approaches may have issues due to the structure of common hash functions like SHA-256
 - MAC with SHA-1 and SHA-2: possibility of length extension attack
- Hash based MAC (HMAC) is a standardized form and the most common approach, it splits a key in two and hashes twice
 - We hash two times to be safer.
 - We split the key into two and we hash twice with each key
 - Then not vulnerable to length extension attack

7.9. SYMMETRIC CRYPTOGRAPHY

Symmetric key algorithms rely on a shared secret key that is distributed to all members who participate in the communications.

- This key is used by all parties to both encrypt and decrypt messages
- The sender encrypts with the shared secret key and the receiver decrypts with it.
- When large-sized keys are used, symmetric encryption is very difficult to break.

It provides for the security service of confidentiality

7.9.1. Stream ciphers

We can approximate a one-time pad by generating an infinite pseudo-random keystream

- Stream ciphers work on messages of any length
- The nonce guarantees that each keystream is unique, even if the same key is reused

7.9.1.1. Pros and cons

<i>Pros</i>	<i>Cons</i>
Encryption of long continuous streams, possibly of unknown length	The keystream must appear statistically random
Extremely fast with a low memory footprint, ideal for low-power devices	You must never reuse a key + nonce
If designed well, it can seek to any location in the stream	Stream ciphers do not protect the ciphertext (= no guaranteed integrity)

7.9.2. Block ciphers

Block ciphers take an input of a fixed size and return an output of the same size

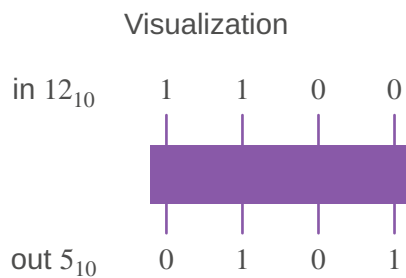
- Block ciphers attempt to hide the transformation from message to ciphertext through confusion and diffusion
- Most block ciphers are SP-Networks

The Advanced Encryption Standard (AES) is an SP-Network

- Almost everything uses AES
- There are others (e.g. Feistel Ciphers)

7.9.2.1. S-Box

Provides Confusion

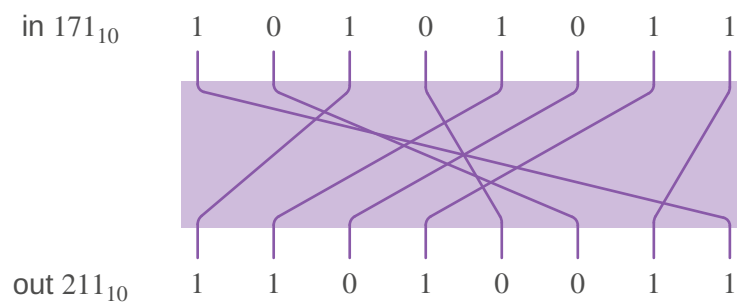


Lookup Table

<i>in</i>	0	1	...	12	...	15
<i>out</i>	6	13	...	5	...	4

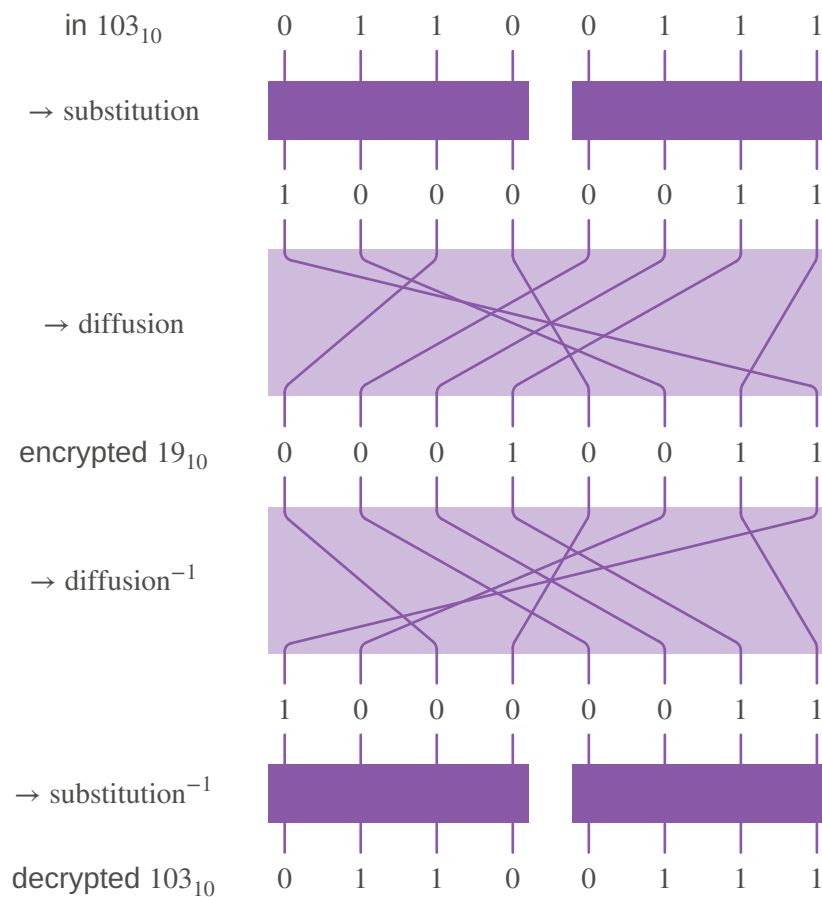
7.9.2.2. P-Box

Provides Diffusion



7.9.2.3. Encryption and decryption in a basic SP-Network

Combines S-Box and P-Box

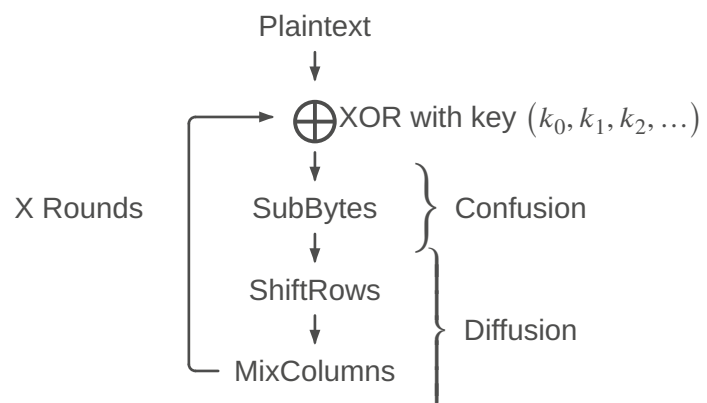


7.9.2.4. Advanced Encryption Standard (AES)

- A standard built around the Rijndael algorithm
- Superseded DES as a standard in 2002
- SP-Network with a 128-bit block size
- Key length of 128, 192 or 256-bits
- 10, 12 or 14 rounds
- Each Round:
 - SubBytes
 - ShiftRows
 - MixColumns
 - Key Addition

Round Structure

$a_{0,0}$	$a_{0,1}$	$a_{0,2}$	$a_{0,3}$
$a_{1,0}$	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$
$a_{2,0}$	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$
$a_{3,0}$	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$



Key addition to a Block / XOR

128 bits block after XOR with the extended key

b_{00}	b_{01}	b_{02}	b_{03}	b_{04}	b_{05}	b_{06}	b_{07}	b_{08}	b_{09}	b_{10}	b_{11}	b_{12}	b_{13}	b_{14}	b_{15}
----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

b_{00}	b_{04}	b_{08}	b_{12}
b_{01}	b_{05}	b_{09}	b_{13}
b_{02}	b_{06}	b_{10}	b_{14}
b_{03}	b_{07}	b_{11}	b_{15}

SubBytes

It is a lookup table, there is no fixed point (byte 15 doesn't end up byte 15)

There is no opposite bit flap (10101010 didn't become 01010101)

$S(b_{00})$	$S(b_{01})$	$S(b_{02})$	$S(b_{03})$	$S(b_{04})$	$S(b_{05})$	$S(b_{06})$	$S(b_{07})$	$S(b_{08})$	$S(b_{09})$	$S(b_{10})$	$S(b_{11})$	$S(b_{12})$	$S(b_{13})$	$S(b_{14})$	$S(b_{15})$
-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------

$S(b_{00})$	$S(b_{04})$	$S(b_{08})$	$S(b_{12})$
$S(b_{01})$	$S(b_{05})$	$S(b_{09})$	$S(b_{13})$
$S(b_{02})$	$S(b_{06})$	$S(b_{10})$	$S(b_{14})$
$S(b_{03})$	$S(b_{07})$	$S(b_{11})$	$S(b_{15})$

ShiftRows

$S(b_{00})$	$S(b_{05})$	$S(b_{10})$	$S(b_{15})$	$S(b_{04})$	$S(b_{09})$	$S(b_{14})$	$S(b_{03})$	$S(b_{08})$	$S(b_{13})$	$S(b_{02})$	$S(b_{07})$	$S(b_{12})$	$S(b_{01})$	$S(b_{06})$	$S(b_{11})$
-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------

$S(b_{00})$	$S(b_{04})$	$S(b_{08})$	$S(b_{12})$	No changes
$S(b_{05})$	$S(b_{09})$	$S(b_{13})$	$S(b_{01})$	1 to the left
$S(b_{10})$	$S(b_{14})$	$S(b_{02})$	$S(b_{06})$	2 to the left
$S(b_{15})$	$S(b_{03})$	$S(b_{07})$	$S(b_{11})$	3 to the left

MixColumns

MixColumns is done using a matrix multiplication

– Add operation is an XOR

- Multiplication operation is a multiplication within that finite field (modular polynomial)

$$\begin{pmatrix} 2 & 3 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} b_{00} \\ b_{05} \\ b_{10} \\ b_{15} \end{pmatrix} = \begin{pmatrix} 2b_{00} + 3b_{05} + b_{10} + b_{15} \\ b_{00} + 2b_{05} + 3b_{10} + b_{15} \\ b_{00} + b_{05} + 2b_{10} + 3b_{15} \\ 3b_{00} + b_{05} + b_{10} + 2b_{15} \end{pmatrix}$$

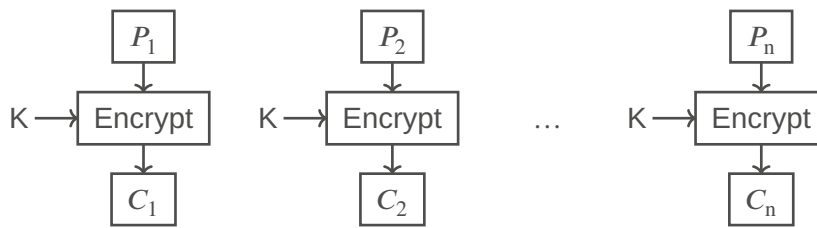
7.9.2.5. Modes of operation for block ciphers

- Realistically, messages of exactly 128-bits are pretty unlikely
 - We need some mechanism to encrypt messages that are longer or shorter
- A mode of operation is the combination of multiple instances of block encryption into a usable protocol
- There are several modes of operations, in this lecture we only cover the following:
 - Electronic Code Book (ECB)
 - Cipher Block Chaining (CBC)
 - Counter Mode (CTR)

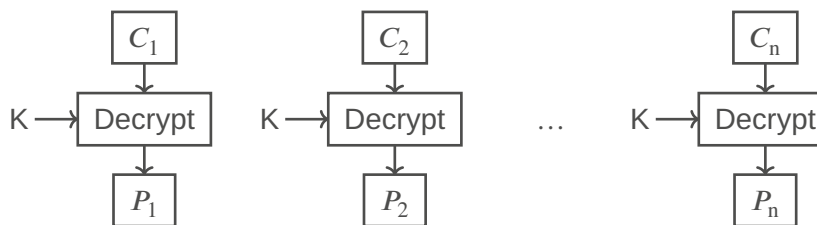
7.9.2.6. Electronic Code Block (ECB)

- Just encrypt each block one after another with same key
- Weak to redundant data divulging patterns
- Electronic codebook is not recommended!

Encryption



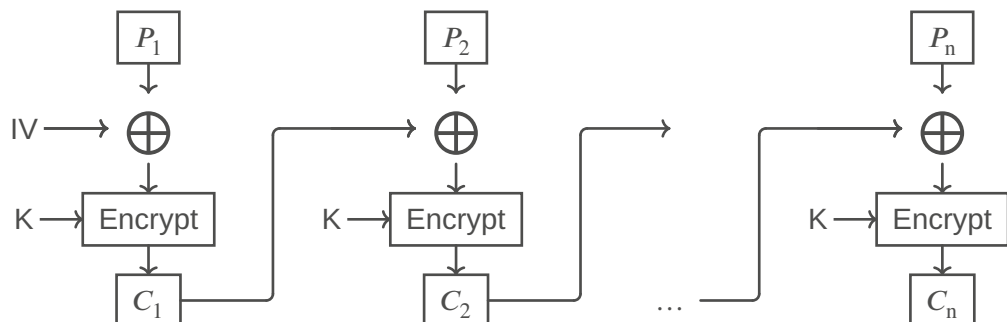
Decryption

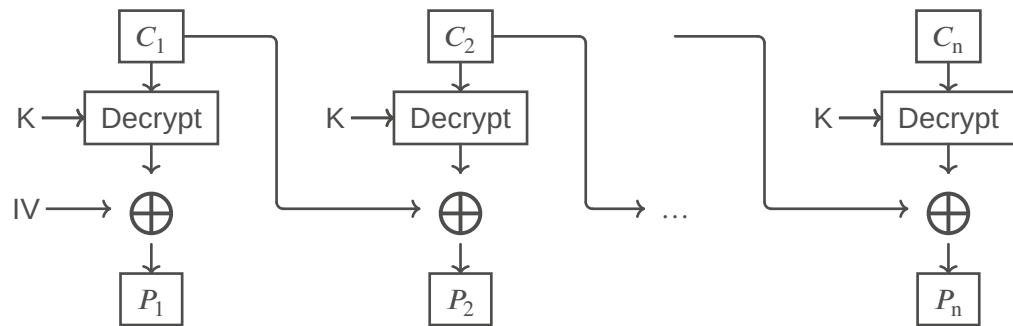


7.9.2.7. Cipher Block Chaining (CBC)

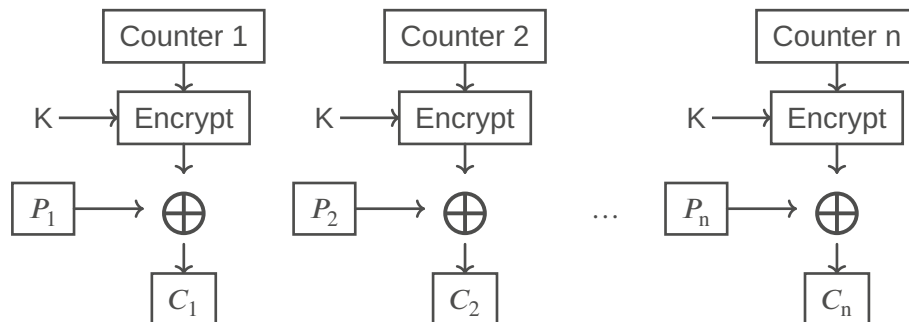
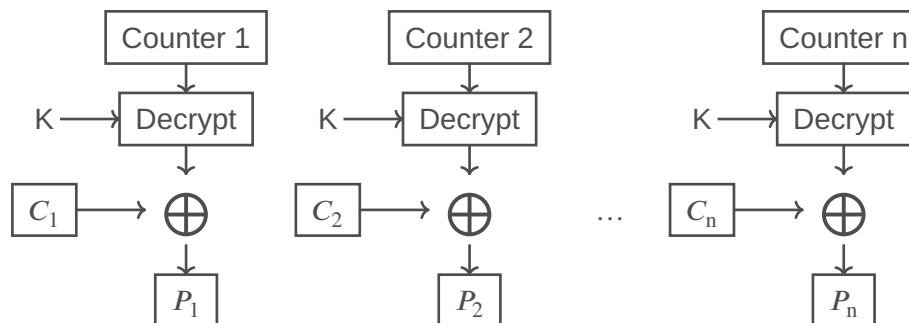
- XOR the IV with the first input, then XOR the output of each cipher block with the next input
 - Not parallelizable
 - It is better than ECB but not perfect

Encryption



Decryption**7.9.2.8. Counter Mode (CTR)**

- Encrypting a counter to produce a stream cipher
 - Pretty good - can also be parallelized!
 - Convert a block cipher into a stream
- We don't encrypt the message
 - We encrypt a number and use the random number that comes out to XOR the message
- Standard mode for all type of encryption cipher (AES)

Encryption**Decryption****7.9.3. Remarks**

Key distribution

- Parties must have a secure method of exchanging the secret key before establishing communications with a symmetric key protocol

Symmetric key cryptography does not implement non-repudiation

- Because any communicating party can encrypt and decrypt messages with the shared secret key, there is no way to prove where a given message originated

Symmetric key cryptography does not implement message integrity

The major strength of symmetric key cryptography is the great speed at which it can operate

- Symmetric key encryption is very fast, often 1,000 to 10,000 times faster than asymmetric algorithms
- Lots of the processor have an AES instruction set

– Alternative to AES: the Chacha20 cipher

TODO:

Diffie hellman, RSA, Elliptic-Curve Diffie Hellman (ECDH)